

16 Set up flags

Add two flag variables to the code to keep track of whether Screen Pet's eyes are crossed or its tongue is out. Type them just above the line that tells Screen Pet to start blinking, which you added to the main part of the code in Step 9.

```
c.eyes_crossed = False
c.tongue_out = False

root.after(1000, blink)
```

These are the flag variables for the pupils and the tongue.

17 Toggle the tongue

This function toggles Screen Pet's tongue between being out and in. Put the code shown below above the `show_happy()` function that you created in Step 11.

```
def toggle_tongue():
    if not c.tongue_out:
        c.itemconfigure(tongue_tip, state=NORMAL)
        c.itemconfigure(tongue_main, state=NORMAL)
        c.tongue_out = True
    else:
        c.itemconfigure(tongue_tip, state=HIDDEN)
        c.itemconfigure(tongue_main, state=HIDDEN)
        c.tongue_out = False
```

```
def show_happy(event):
```

This line sets a flag variable saying the tongue isn't out.

The code checks to see if the tongue is out already.

If the tongue isn't out, these lines make it visible.

This line sets a flag variable saying the tongue is now out.

The tongue is already out (else).

These lines hide the tongue again.

EXPERT TIPS

Using flag variables

Flag variables help you keep track of something in your program that can be in one of two states. When you change the state, you update the flag. The "Engaged / Vacant" sign on a toilet door is a flag – you set it to "Engaged" when you lock the door and back to "Vacant" when you unlock it.

